

(SW) also use given

$$1) E(x) = 1.5 \quad E(3x) = 4.5$$

$$2) \sum P(x) = k^2 - 48 \Rightarrow k^2 - 48 = 1$$

$$k^2 = 49 \quad \therefore k = \pm 7$$

$$3) E(x) - E(E(x)) = E(x) - E(x) = 0$$

$$4) \sum F(x) = 1 \Rightarrow 5k + 4k + 3k + k = 1$$

$$15k = 1 \quad \therefore k = \frac{1}{15}$$

$$5) V(3x) + V(4y) + 2\text{Cov}(3x, 4y)$$

$$= 3^2 V(x) + 4^2 V(y) + 0$$

$$= 3^2 \cdot 0.2 + 0.5 \cdot 4^2 = 9.8$$